Master's Degree Proposals

Lattice-based Folding Schemes

Background. Classical elliptic curve-based cryptography enabled numerous protocols and is arguably the cryptography branch that yields the most efficient schemes (signature schemes such as Schnorr or ECDSA with sizes below 100 B, succinct proving systems based on elliptic curve pairing etc.) However, such schemes are not secure against quantum computers. Consequently, the question arises how to construct post-quantum secure counterparts of previous schemes built on top of elliptic curves.

Currently lattice-based cryptography is the most promising solution to this problem. Specifically, one of the major lattice-based assumptions (called *Short Integer Solution*) states that finding the "short" solution to equation $\mathbf{Ax} = 0$ for a random matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ with respect to $\mathbf{x} \in \mathbb{Z}_q^m$ is hard (where by "short" we mean $\|\mathbf{x}\|_\infty < \gamma$ for appropriate γ). Many major tricks to making such schemes more efficient is related to considering the cyclotomic number field $\mathbb{K} := \mathbb{Q}[X]/(X^n + 1)$ for $n = 2^k$, and using R/qR instead of $\mathbb{Z}_q$ (where $R \cong \mathbb{Z}[X]/(X^n + 1)$ is the ring of integers of $\mathbb{K}$). That said, the equation $\mathbf{Ax} = 0$ is now solved for a random matrix in $R_q^{n \times m}$ and the norm is now taken with respect to the geometry inherited from the structure of $\mathbb{K}$ (such modification is called *Module Short Integer Solution* or MSIS for short).

Goal. There are currently numerous studies on how to apply such assumptions on "classical" cryptographical constructions (such as digital signatures or public key encryption). However, it is still unresolved how to put more advanced protocols on lattice-based rails: multisignatures, threshold signatures, succinct proving systems etc.

The promising line of research is *folding schemes*. Recall that the goal of proving systems over binary relation $\mathscr{R}$ is to demonstrate the knowledge of the secret data (witness) $\mathbb{w}$ such that $(\mathbb{x}, \mathbb{w}) \in \mathscr{R}$ for a public statement $\mathbb{x}$. The *reduction* $\Pi : \mathscr{R} \to \mathscr{R}'$ is the protocol allowing to reduce proving the statement $\mathbb{x}$ over $\mathscr{R}$ to proving the statement $\mathbb{x}'$ over $\mathscr{R}'$ (with the property that any efficient witness extractor of $\mathscr{R}'$ can extract the witness from $\mathscr{R}$). Finally, the *folding scheme* is a reduction $\Pi_{\text{fd}} : \mathscr{R}_{\text{agg}} \times \mathscr{R}_{\text{comp}} \to \mathscr{R}_{\text{agg}}$: i.e., instead of considering a complicated relation $\mathscr{R}_{\text{agg}} \times \mathscr{R}_{\text{comp}}$ we reduce it to a single relation $\mathscr{R}_{\text{agg}}$. This allows to fold proving thousands of same relations to a single relation $\mathscr{R}_{\text{agg}}$.

Turns out that lattice assumptions allow to build effective folding schemes. Most recent papers on these topics are *LatticeFold* [BC24] by Dan Boneh and Binyi Chen (Stanford), *Neo* [NS25] by Wilson Nguyen and Srinath Setty (Stanford and Microsoft Research), and *Cyclo* [GLLO26] by Albert Garreta et al. (University of Tartu and Nethermind).

Goal #1. Survey all aforementioned schemes and understand underlying techniques.

Goal #2. The schemes presented are *not zero-knowledge*. The goal is to build the zero-knowledge-secure version of one of presented protocols and prove it.

Details. Two primary reasons why they are not zero-knowledge:

- Underlying sum-check protocol (*i.e.*, showing $\sum_{\vec{\mathbf{b}} \in \{0,1\}^\nu} f(\vec{\mathbf{b}}) = H$ for $f \in R[X_1, \ldots, X_\nu]$ and public $H \in R$ for some ring R) is not zero-knowledge by default. One should apply techniques to make it zero-knowledge (*e.g.*, apply techniques from [CFS17]).
- Ajtai commitment scheme $\mathbf{x} \mapsto \mathbf{Ax}$ is *binding* under MSIS assumption, but not hiding. Is it possible to use randomized (hiding) Ajtai commitment scheme $(\mathbf{x}, \mathbf{r}) \mapsto \mathbf{Ax} + \mathbf{Br}$ instead? How much worse schemes become in terms of efficiency?

Goal #3. One concrete practical application of folding scheme is batching multiple *range proofs* (proofs that some private amount $s \in \mathbb{N}$ lies in range $0 \leq s \leq L$ for $L \in \mathbb{N}$). Is it possible to specifically optimize the scheme where $\mathscr{R}_{\text{comp}}$ is the range proof?

Possible solution. Apply techniques from [ALS20] and see what happens.

Isogeny-based Cryptography

Another quite novel area of post-quantum research requiring arguably the most heavy algebraic preliminaries is *isogeny-based cryptography*.

Goal #1. Understand the SQISign construction:

`https://sqisign.org/spec/sqisign-20250707.pdf`

Goal #2. Is it possible to construct key derivation, multisignatures, threshold signatures on top?

Bibliography

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